

STATISTICAL ARBITRAGE · 2026

Modern Pairs Trading

Ten Techniques with Formulas, Procedures and Worked Numbers

US large caps and ETFs. Each technique comes with its procedure and computed results.

Selection

Dynamics

Timing

Validation

All examples use simulated data — read them as comparisons, not absolute levels

Overview

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EXECUTIVE SUMMARY

The progress of recent techniques is in execution, not alpha

What changed from the classical method, split into four links.

Problem

Classical pair returns shrank through competition, costs and unstable relations

Fix

Reinforce each of the four links: selection · dynamics · timing · validation

Gain

Clustering: false positives 61→11; Kalman: residual sd ÷12; optimal band: profit rate ×1.6–2.1

Left over

Copulas and deep learning did not beat the z-score in these examples

The point is not finding new alpha but not wasting the signal you already have

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Why it weakened

Competition · costs · unstable relations

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⑦ Optimal boundaries ⑧ RL ⑨ Deep learning

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Size of gains and order of adoption

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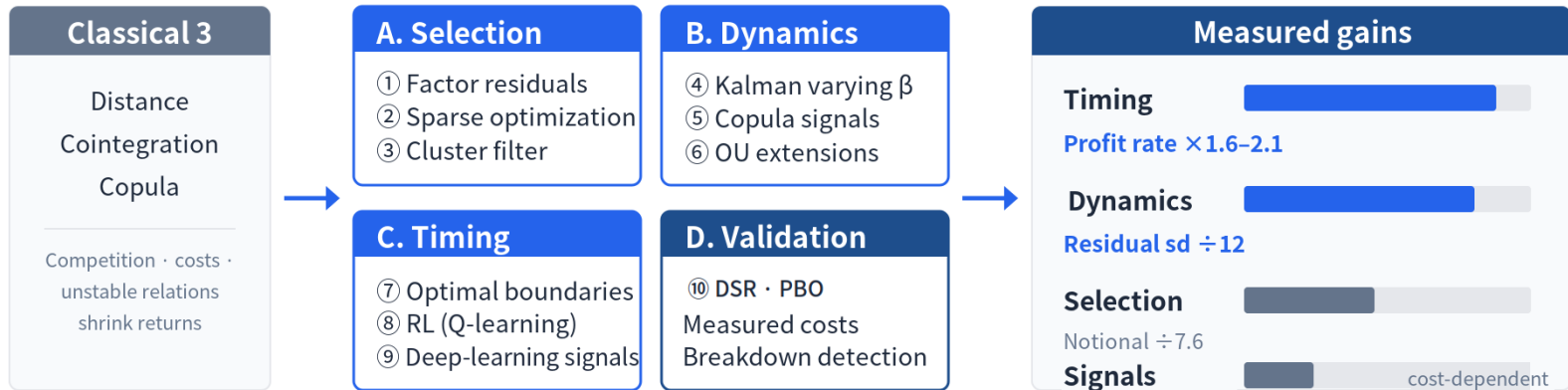
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01. Why it weakened

Recent techniques are placed on four weak links

Left: classical; centre: where it is reinforced; right: the measured gain.



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01. Why it weakened

Returns shrank because three things happened at once

Competition**A known rule
fades as it becomes known**

Distance-method excess returns come from 1962–2002; since 2003 they have more than halved.

Gatev et al. 2006

Costs**A round trip needs
four executions**

Tighter bands turn over faster and costs eat the return. Gross and net diverge.

4 executions + borrow

Unstable relations**The formation β
is gone when trading**

Restructuring and index changes break cointegration; a fixed β cannot absorb it.

Structural change

So the goal is not a stronger signal but executing the same signal more cheaply and stably

02

Selection

What to trade

Factor residuals · sparse mean reversion · cluster filter

02. Selection · Technique ①

The residual left after removing factors is what you trade

Decompose the correlation matrix; the top K principal components are the factors.

$$R_{it} = \alpha_i + \sum_{k=1}^K \beta_{ik} F_{kt} + \epsilon_{it}, \quad X_{it} = \sum_{\tau \leq t} \epsilon_{i\tau}$$

$$s_{it} = \frac{X_{it} - m_i}{\sigma_{\text{eq},i}}, \quad \kappa_i = -\ln b_i, \quad \sigma_{\text{eq},i} = \sqrt{\frac{\text{Var}(\zeta_i)}{1 - b_i^2}}$$

- **Standardize + PCA** Decompose correlation, not covariance, so volatility does not dominate
- **Choosing K** 55–65% cumulative variance or an eigenvalue gap. Re-select each window
- **OU fit · filter** Get the half-life from an AR(1); drop names above 30 days

Enter when the s-score exceeds 1.25; exit when it falls below 0.50

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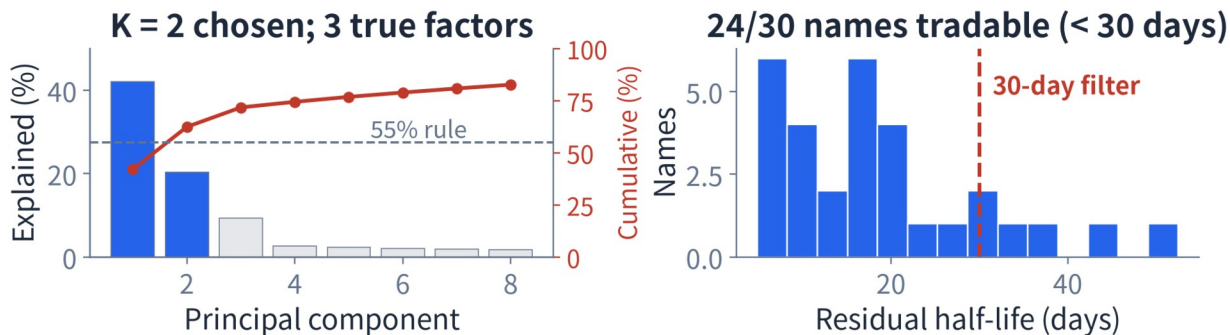
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02. Selection · Technique ①

Choose K wrong and a factor stays in the residual

30 names × 500 days from a 3-factor structure. Loadings 1.0 / 0.6 / 0.4.

**42.0%**

PC1
explained

K = 2

55% rule
true factors: 3

24/30

tradable
half-life < 30 days

53

entry signals
last 60 days

02. Selection · Technique ①

The third factor stays in the residual and mixes in common risk

Cost of choosing K too small

- The third factor (loading 0.4) explains only 9.3%
- The 55% rule cuts at $K = 2$
- Part of the residual is really factor exposure

Cost of choosing K too large

- Tradable residual variation disappears
- A 75% rule raises K to 5
- Fewer signals, larger estimation error

- **In practice** Re-select K each window instead of fixing it, and log the number of tries

The most common way common risk creeps into a position believed market-neutral

02. Selection · Technique ②

Maximize reversion speed with a cap on names

Johansen controls neither the vector's size nor the number of legs.

$$\max_w \kappa(w) \quad \text{s.t.} \quad \|w\|_0 \leq k, \quad \|w\|_2 = 1$$

$$\text{cost ratio} = \frac{2\tau \sum_i |w_i| P_i}{\sigma_{\text{spread}}}$$

- **Shrink the universe** Split the candidates by clustering first
- **Fix the rank** Confirm the number of cointegrating relations r with Johansen
- **Cardinality search** Compute the half-life of each subset with $k = 2, 3, 4$
- **Compare after costs** Look at notional and cost ratio, not just the half-life

An ℓ_0 constraint makes it combinatorial; with many names, approximate by greedy search or annealing

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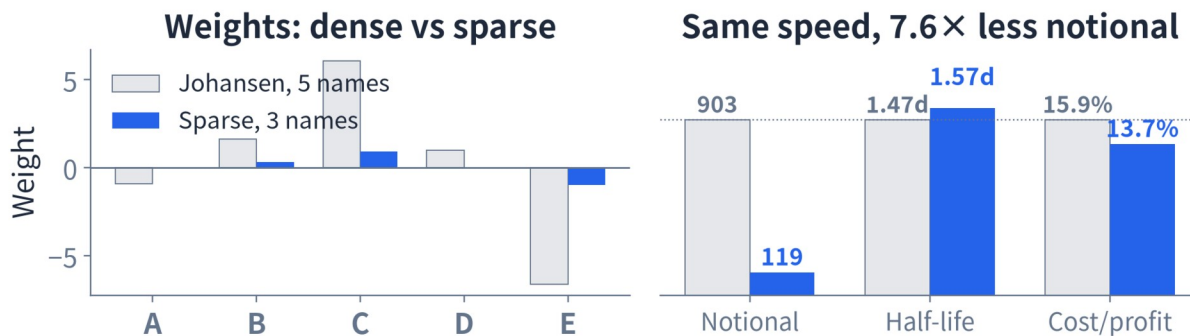
Validation

Synthesis

02. Selection · Technique ②

The same reversion speed with 7.6× less notional

5 assets from 3 common trends, $T = 800$, 5bp one way.

**903**

Johansen notional
5 legs

119

Sparse notional
3 legs

1.47 / 1.57

Half-life (days)
effectively equal

15.9 → 13.7

Cost ratio (%)
2.2pp better

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02. Selection · Technique ②

Sparsity pays off in execution risk more than in cost

What actually improves

- 5 → 3 legs: less partial-fill risk
- Notional 903 → 119: less borrowing, margin
- Weights 6.6 → 1.0: less error amplification

What not to overstate

- The cost-ratio gain is only 2.2pp
- Sparse sd 0.87, one-sixth of Johansen's
- Small absolute profits may not clear costs

- **Check together** Half-life alone as the objective picks untradable spreads

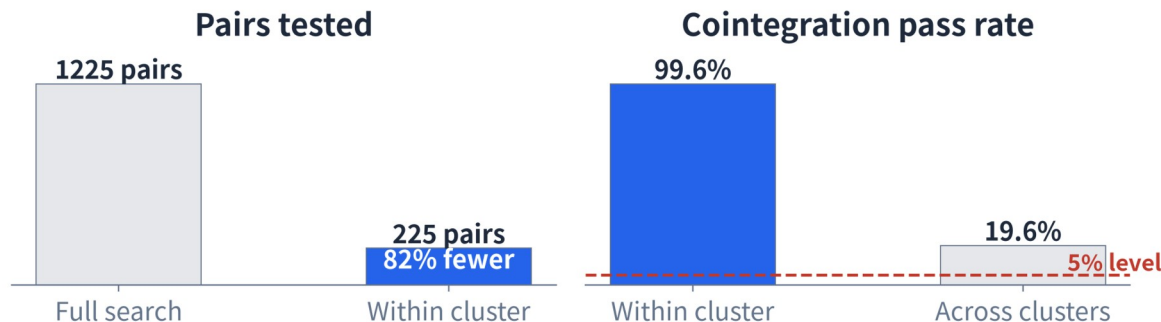
Only reversion speed and volatility together give a basket you can actually trade

02. Selection · Technique ③

Clustering cuts the multiple-testing burden first

Turn correlation into distance and cluster hierarchically.

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}, \quad \mathbb{E}[\text{false}] = \alpha \binom{N}{2}$$



82% fewer candidates; expected false positives at 5% fall from 61 to 11

* Simulated 50 names × 600 days, 5-cluster structure. Cluster purity 1.00

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02. Selection · Technique ③

A 19.6% cross-cluster pass rate exposes false positives

Full search

Within-cluster search

✗ Test all 1,225 pairs



✓ Test only 225 pairs

✗ 61 expected false positives



✓ 11 expected false positives

✗ Many passing pairs are chance



✓ Pairs with an economic rationale

Caveat

Use clusters only as a filter; leave the final pick to the cointegration test. Include a stress period and everything falls into one cluster.

03

Dynamics

How to track the relation

Kalman time-varying beta · copula signals · regime-change detection

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03. Dynamics · Technique ④**Treat beta as a state variable, not a constant**

The one-step forecast error is the traded spread, standardized by its forecast variance.

$$e_t = Y_t - F_t \theta_{t-1}, \quad S_t = F_t P_{t|t-1} F_t' + V_e, \quad z_t = \frac{e_t}{\sqrt{S_t}}$$

$$K_t = \frac{P_{t|t-1} F_t'}{S_t}, \quad \theta_t = \theta_{t-1} + K_t e_t, \quad V_w = \frac{\delta}{1-\delta} I$$

- **Initialize · warm up** Set $\theta = (0, 1)$, $P = I$ and discard the first 100 periods
- **Choosing δ** Grid 10^{-5} – 10^{-3} . Watch the residual ADF p-value and sd together
- **Check stationarity** ADF after the warm-up. If it fails to reject, choose δ again

Choosing δ counts as trials too — it is corrected for in validation

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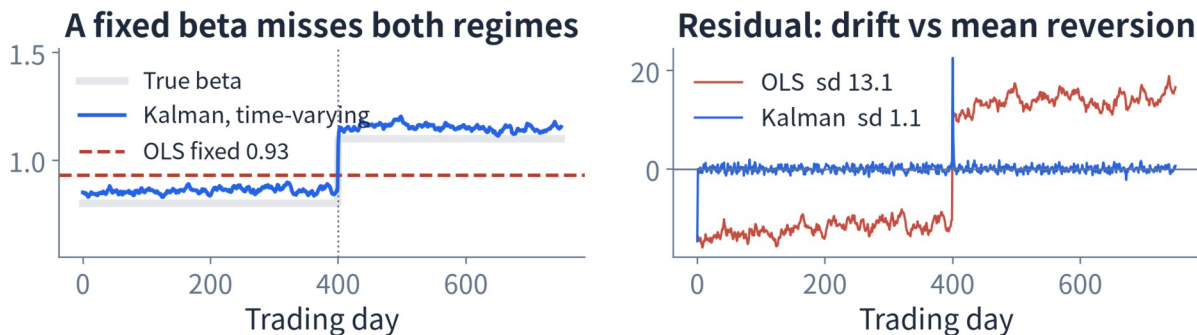
Validation

Synthesis

03. Dynamics · Technique ④

When the relation shifts, a fixed beta loses stationarity

Simulation: the true β shifts from 0.80 to 1.10 at $t = 400$.

**0.930**

OLS fixed β
between regimes

13.07

OLS residual sd
off in a step

1.10

Kalman residual sd
cut to one-twelfth

0.772

OLS ADF p
unit root not rejected

0.000

Kalman ADF p
unit root rejected

Overview

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03. Dynamics · Technique ④

Trade the OLS residual and you trade the regime gap

Errors a fixed β creates

- Buy signal on through the first half
- Sell signal on through the second half
- It trades the regime gap, not mean reversion

Tuning δ is a balance

- Large δ : β chases noise too
- Small δ : late to a regime shift
- Good at gradual shifts, still late at abrupt breaks

- **Complement** Watch abrupt breaks with the regime detection of technique ⑥

Keep the warm-up and the initial convergence creates false signals

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03. Dynamics · Technique ⑥

A copula measures unusualness, not the size of the deviation

Turn the marginals into ranks and judge by conditional cumulative probability.

$$h_{1|2}(u_1 \mid u_2) = \frac{\partial C(u_1, u_2)}{\partial u_2} = \Phi \left(\frac{x_1 - \rho x_2}{\sqrt{1 - \rho^2}} \right)$$

- **Rank transform** Rank the log returns to get u
- **Measure dependence** Compute Kendall's tau, free of the marginals
- **Choose the family** Fit Gaussian, Clayton, Gumbel and t ; pick by AIC
- **Two-sided check** Enter only when the two directions point opposite ways

Strong lower-tail dependence: Clayton; upper tail: Gumbel

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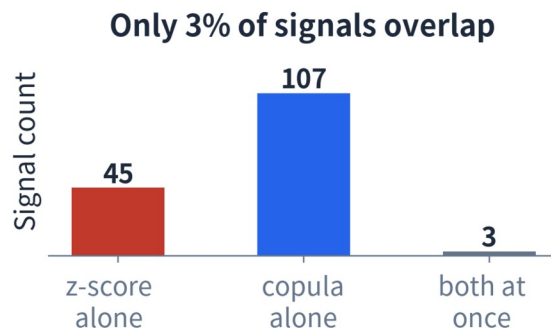
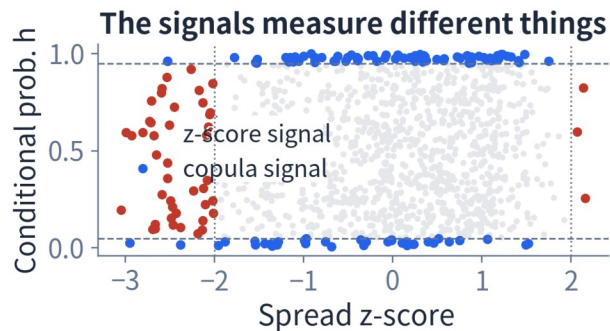
Validation

Synthesis

03. Dynamics · Technique ⑥

The two signals coincide only 3% of the time

A pair with joint downside shocks, $T = 1200$, Kendall's tau 0.340.

**45**

z-score alone
 $|z| > 2$

107

copula alone
 $h < 0.05$ or > 0.95

3

both at once
3% of copula signals

0.432 / 0.484

downside / upside h
downside more unusual

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03. Dynamics · Technique ⑥

2.4× the trades means 2.4× the cost

Item	z-score	copula h	Reading
Measures	deviation of the cumulative spread	unusualness of today's return pair	different things
Distribution	normal approximation	marginals and dependence separated	captures tail asymmetry
Signal count	48	110	2.4×
Empirical results	benchmark	no clear risk-adjusted edge	Rad et al. 2016

Caveat

At equal signal quality, more trades make net returns worse. Remember the choice of family is itself a degree of freedom.

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03. Dynamics · Technique ©

Watch for regime change with a rolling half-life

The formal estimator is an HMM; in practice a rolling estimate serves as the proxy.

$$dX_t = -\kappa_{s_t} X_t dt + \sigma_{s_t} dW_t, \quad \widehat{HL}_t = \frac{\ln 2}{-\ln \hat{b}_t}$$

- **Normal baseline** Take the median rolling half-life over a calm period
- **Window length** Start at 15–20× the half-life. Too short is noisy, too long is late
- **Alarm rule** Alarm when it exceeds 2× the normal median
- **Response** Stop new entries; widen the bands on open positions

Rolling estimates understate the half-life — watch the direction of change, not the level

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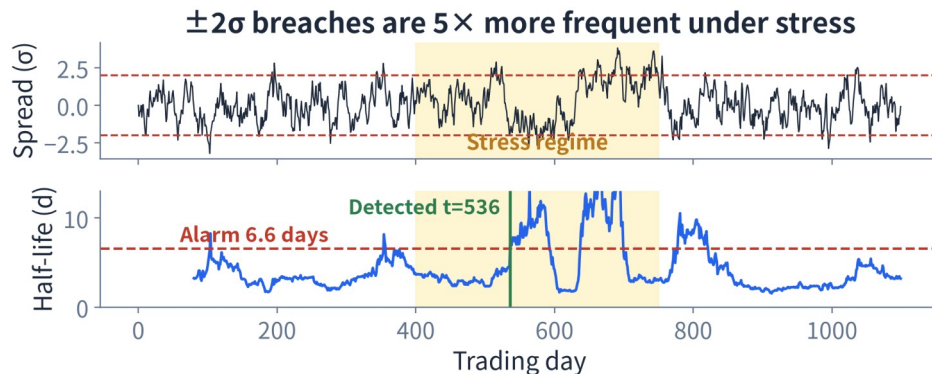
Validation

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03. Dynamics · Technique ⑥

Detection works, but with long lags and large bias

From $t = 400$, the half-life switches from 4 to 22 days for 350 days. Rolling window 80 days.

**3.3 → 7.3**

Rolling estimate (days)

true 4.0 → 22.0

136 days

detection lag

1.7× the window

4% → 22%share beyond $\pm 2\sigma$

5× more

4 → 13 days

longest run outside

chained stop-losses

04

Timing

When to enter and exit

Optimal double stopping · reinforcement learning · deep-learning signals

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04. Timing · Technique ②

 $\pm 2\sigma$ is a convention, not the result of any optimization

Treat entry and exit as stopping times and maximize profit per unit time.

$$\max_b \frac{b-c}{\mathbb{E}[T_{-b \rightarrow 0}] + \mathbb{E}[T_{0 \rightarrow -b}]}$$

$$\mathbb{E}[T_{a \rightarrow b}] = \frac{2}{\sigma^2} \int_a^b e^{\kappa y^2 / \sigma^2} \left(\int_{-\infty}^y e^{-\kappa z^2 / \sigma^2} dz \right) dy$$

- **Parameters** Get κ and σ_{eq} from an AR(1)
- **Price the costs** Convert effective spread, fees, borrow and impact into σ units
- **Grid search** Scan b from 0.1σ to 3σ in steps of 0.05

Leung & Li (2015) — add a stop-loss and the entry region becomes a bounded interval

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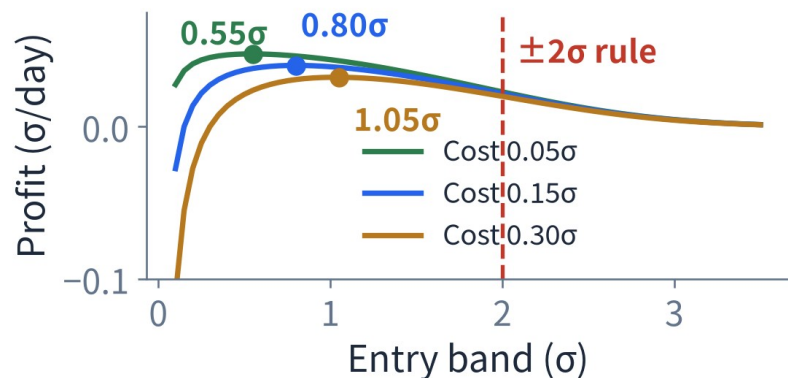
Validation

Synthesis

04. Timing · Technique ⑦

Costs, not volatility, set the band

Assume an OU spread with a 5-day half-life and stationary sd of 1σ .



Higher costs widen the band

- Cost 0.05 σ $\rightarrow$ $b \cdot 0.55\sigma$ · cycle 10.5 days
- Cost 0.15 σ $\rightarrow$ $b \cdot 0.80\sigma$ · cycle 16.2 days
- Cost 0.30 σ $\rightarrow$ $b \cdot 1.05\sigma$ · cycle 23.1 days
- Conventional $\pm 2\sigma$ $\rightarrow$ about half the profit rate
- 1.63–2.09 $\times$ the conventional band

With zero cost the optimal band shrinks to zero — the band is a product of costs

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04. Timing · Technique ⑧

Learn the boundary instead of solving for it

State: standardized spread and position. Action: target position.

$$R_t = \pi_{t-1}(X_{t+1} - X_t) - |\pi_t - \pi_{t-1}| \cdot \frac{c}{2}$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha [R_t + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)]$$

- **Discretize the state** z in 20 bins over -2.5 to 2.5 , times 3 positions
- **Reward design** Always put costs in the reward, or it converges to overtrading
- **Benchmark** Compare with the analytic optimal band and the conventional band out of sample
- **Symmetry check** Asymmetric boundaries in a symmetric setting are learning noise

Exploration rate ϵ decays linearly from 0.35 to 0.02

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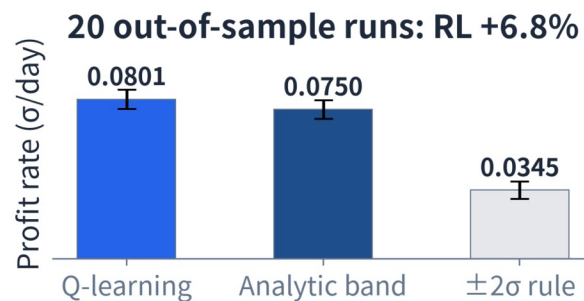
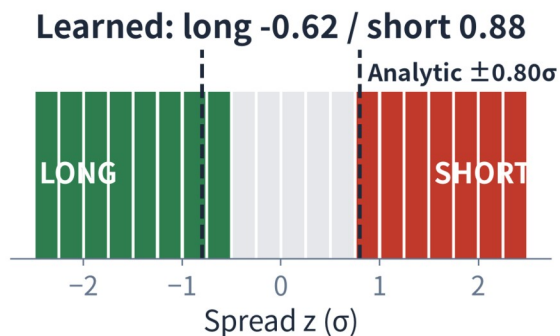
Validation

Synthesis

04. Timing · Technique ⑧

In an ideal setting, RL beats the analytic solution by 6.8%

Half-life 5 days, round-trip cost 0.15σ . 60 training episodes, then 20 out-of-sample runs.

**0.0801**

Q-learning
out-of-sample mean

0.0750

Analytic band
 $b^* = 0.80\sigma$

0.0345

Conventional $\pm 2\sigma$
less than half

95%

RL win rate
19 of 20 runs

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04. Timing · Technique ⑧

RL's edge comes from two conditions

Reason ① the benchmark's constraint

- A band rule must re-cross the band to re-enter
- RL picks the best action in every state
- Analytic: best band rule, not best overall

Reason ② an ideal setting

- OU parameters fixed, state fully observed
- Trained on 240,000 simulated steps
- 20 years of daily data is only 5,000 days

- **Learning noise** The setting is symmetric, yet the learned boundaries are asymmetric: $-0.62 / +0.88$

RL pays where no analytic solution exists — multi-pair allocation, regime-aware bounds

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04. Timing · Technique ⑨

Adopt deep learning only if it beats the benchmark

Take an L-day residual path and predict the change over the next H days.

$$\hat{y}_t = f_{\theta}(X_{t-L+1}, \dots, X_t), \quad y_t = X_{t+H} - X_t$$

$$\text{IC} = \text{corr}(\hat{y}, y), \quad \text{net} = \overline{\text{sign}(\hat{y}) \cdot y} - c \overline{|\Delta \text{sign}(\hat{y})|}$$

- **Fix the benchmark** Measure the IC and net return of the z-score alone first
- **Split by time** Train, validate, test in time order. A random split leaks information
- **Linear first** If ridge cannot beat the z-score, a nonlinear model usually cannot either
- **Adoption rule** Adopt only if it beats the benchmark at every cost level

A high IC with high turnover still nets little — watch both

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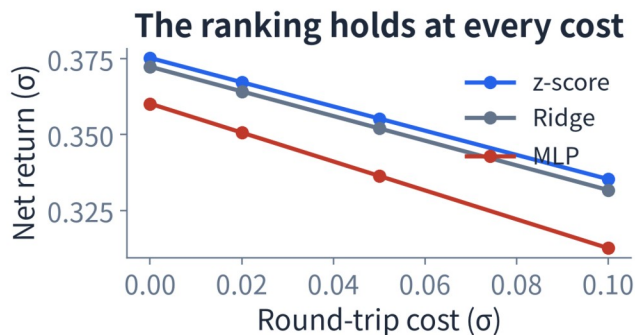
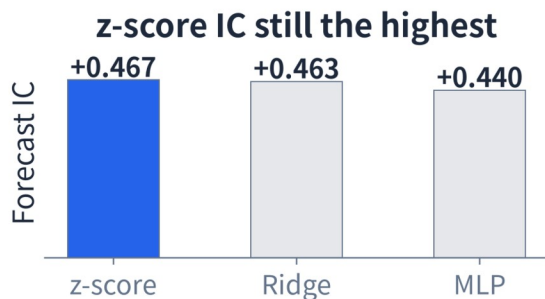
Validation

Synthesis

04. Timing · Technique ⑨

Even in a nonlinear setting, the MLP did not beat the z-score

A nonlinear OU where larger deviations revert faster. Lookback 20 days, horizon 5 days.

**+0.467**

z-score IC
highest

+0.463

Ridge IC
linear info used up

+0.440

MLP IC
actually lower

0.3753

z-score net return
at zero cost

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04. Timing · Technique ⑨

Sophistication is worthless unless it changes the decision

The nonlinearity here

$\kappa(x) = \kappa_0 \times (0.5 + 0.9 \times \min(|x|, 2.5))$ larger deviations revert faster

What deep learning caught

- the size (speed) of reversion
- unused by a sign-based strategy

What the z-score already has

- direction of reversion = $-\text{sign}(x)$
- the nonlinearity does not change the sign

A strategy sizing positions by forecast strength could turn out differently

※ The ranking held at every cost level (0 · 0.02 · 0.05 · 0.10σ)

05

Validation

Can you trust the backtest

A stage for filtering out, not improving

05. Validation · Technique ⑩

Raise the hurdle for the number of trials

Report the best of N trials and pure noise yields a high Sharpe.

$$SR_0 = \hat{\sigma}_{SR} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{N_e} \right) \right]$$

$$DSR = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

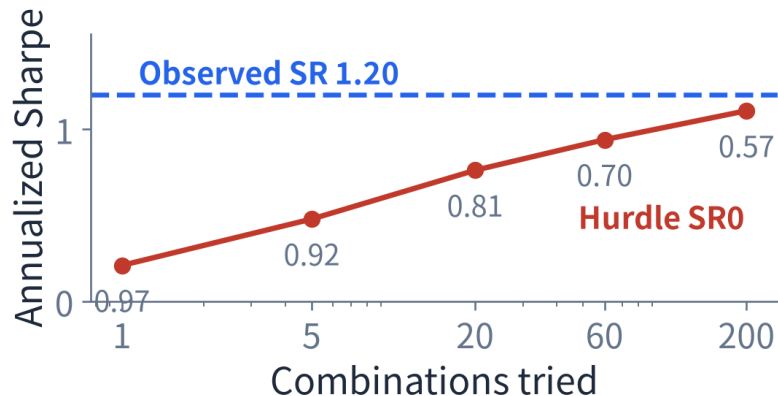
- **Log the trials** Count every change of band, lookback, filter, δ , family and structure
- **Estimate the variance** Take the sd of Sharpe ratios across the tried combinations
- **Decide** Below DSR 0.95, do not adopt without further validation

Enter every Sharpe ratio at the observation frequency (daily)

05. Validation · Technique ⑩

Same Sharpe, different trust — it depends on the search

Observed SR 1.20 annualized, $T = 1000$ days, skew -0.3 , kurtosis 5 .



Trust in the same Sharpe of 1.20

- 1 trial → hurdle 0.21 · DSR 0.974
- 20 trials → hurdle 0.76 · DSR 0.806
- 60 trials → hurdle 0.94 · DSR 0.696
- 200 trials → hurdle 1.11 · DSR 0.573
- At 200 trials it is indistinguishable from a coin toss

This note's own examples are no exception — the δ grid, 59 b values, RL hyperparameters

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05. Validation

A backtest without measured costs is not validation

1 Spread · impact Four executions per round trip; use measured effective spreads per name

2 Borrowing Short-sale availability and stock-borrow fees vary widely by name

3 Funding Include margin and funding for both legs

4 Latency Filling a close signal at the close is optimistic

5 Signal skew Check whether signals cluster in expensive-to-borrow names

Strategies significant gross often vanish net

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05. Validation

Cointegration does not last forever, so set exit criteria

Statistical signal

Rising rolling ADF p-value

If the p-value grows at each re-estimation, the residual's stationarity is weakening

Lengthening half-life

Slower reversion lengthens holding periods and raises costs

Shrinking adjustment speed λ

A smaller $|\lambda|$ in the error-correction model means weaker pull back

More stop-losses hit

More stop-outs at the same band call the relation itself into question

Dynamics signal

Operational signal

When three or more appear together, consider exiting, not re-estimating

06

Synthesis

What actually improved

Numerical conclusions by technique and order of adoption

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Synthesis**06. Synthesis****Numerical conclusions: selection and dynamics**

Technique	What the worked example shows	Gain
① Factor residuals	24/30 names tradable; choosing K is the key	Medium
② Sparse optimization	Same half-life with 7.6× less notional	Medium
③ Cluster filter	82% fewer candidates; expected false positives 61 → 11	Large
④ Time-varying β	Residual sd $\div 12$; ADF rejects	Large
⑤ Copula	2.4× signals, 3% overlap, no clear edge	Small
⑥ Regime detection	136-day lag, large estimation bias	Medium

Multiple-testing control matters most in selection; a time-varying β in dynamics

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Synthesis**06. Synthesis****Numerical conclusions: timing and validation**

Technique	What the worked example shows	Gain
⑦ Optimal boundaries	Profit rate 1.63–2.09× that of $\pm 2\sigma$	Large
⑧ RL	+6.8% over the analytic solution, ideal setting only	Conditional
⑨ Deep learning	Fails to beat the z-score even when nonlinear	None
⑩ DSR	DSR 0.573 after 200 trials	Changes direction

Confirmed real gains

- Cost-aware optimal band
- Multiple-testing control and time-varying beta

Not confirmed

- Refining the signal itself
- Reinforcement learning outside ideal settings

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06. Synthesis

Change the order of adoption and the same effort buys more

- 1 Measure costs** Measure round-trip costs and recompute the optimal band (technique ⑦)
- 2 Cut candidates** Lower the multiple-testing burden with clustering (technique ③)
- 3 Time-varying β** Replace the fixed β with Kalman and recheck stationarity (technique ④)
- 4 Fewer legs** Cut notional and legs through sparsity (technique ②)
- 5 Watch regimes** Set exit criteria in advance (technique ⑥)
- 6 Refine signals** Consider copulas and deep learning last (techniques ⑤ ⑨)

Starting with the sixth step is the most common mistake

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06. Synthesis

What this module leaves out, and next steps

Limits not covered

- Simulated data, not an empirical backtest
- No capital allocation across pairs
- No intraday data or execution models
- HMM regime estimation: concept only

Next steps

- Empirical US ETF lab (yfinance)
- Threshold ECM and cost bands
- Permanent–transitory split, information share
- Pairs Trading in Korea — Korean Stock Pairs and KOSPI200 Cash–Futures

Pairs Trading in Korea shows which techniques short-selling rules and borrow limits disable

**The test is the gate;
what you actually use
is costs and adjustment speed.**

Start by measuring costs, cutting candidates, letting beta move, and counting trials.